



ADVANCED
PARADIGMS IN
ARTIFICIAL
INTELLIGENCE

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CONTRASTIVE LEARNING

For Visual Representations

Group - 7

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OUR OBJECTIVES



1.

LEARN SELF-SUPERVISED VISUAL REPRESENTATIONS

Understand the theoretical foundations and practical motivations behind contrastive learning and explore how models like SimCLR and MoCo learn rich image embeddings without needing labelled data.

2.

IMPLEMENT AND EVALUATE CONTRASTIVE MODELS

Use STL-10 to train SimCLR and later MoCo and BYOL using PyTorch. Apply data augmentations, train with contrastive loss, and evaluate via embedding visualizations and linear probing classifiers.

3.

ANALYZE AND OPTIMIZE PERFORMANCE

Compare contrastive embeddings with those from supervised models on downstream tasks like classification. Perform hyperparameter tuning (e.g., temperature, augmentations) to enhance model performance.



COMPONENTS

DATASET USED

STL-10

- 100K UNLABELED (SSL TRAINING)
- 13K LABELED (EVALUATION)
- 10 CLASSES, 96×96 RESOLUTION

MODELS USED

SIMCLR

- RESNET-18 + MLP HEAD
- NT-XENT LOSS

MOCO

- MOMENTUM ENCODER + QUEUE

BYOL

- ONLINE & TARGET NETWORKS
- NO NEGATIVES USED

FRAMEWORKS AND TOOLS

LANGUAGE: **PYTHON**

DL LIBRARIES:
PYTORCH, TORCHVISION

UTILITIES:
NUMPY, MATPLOTLIB



PROGRESS SO FAR...

1.

UNDERSTANDING & LITERATURE REVIEW

- Studied SimCLR, MoCo, BYOL papers
- Understood key components: augmentations, encoders, loss functions
- Compared design choices: contrastive loss vs. bootstrap methods

2.

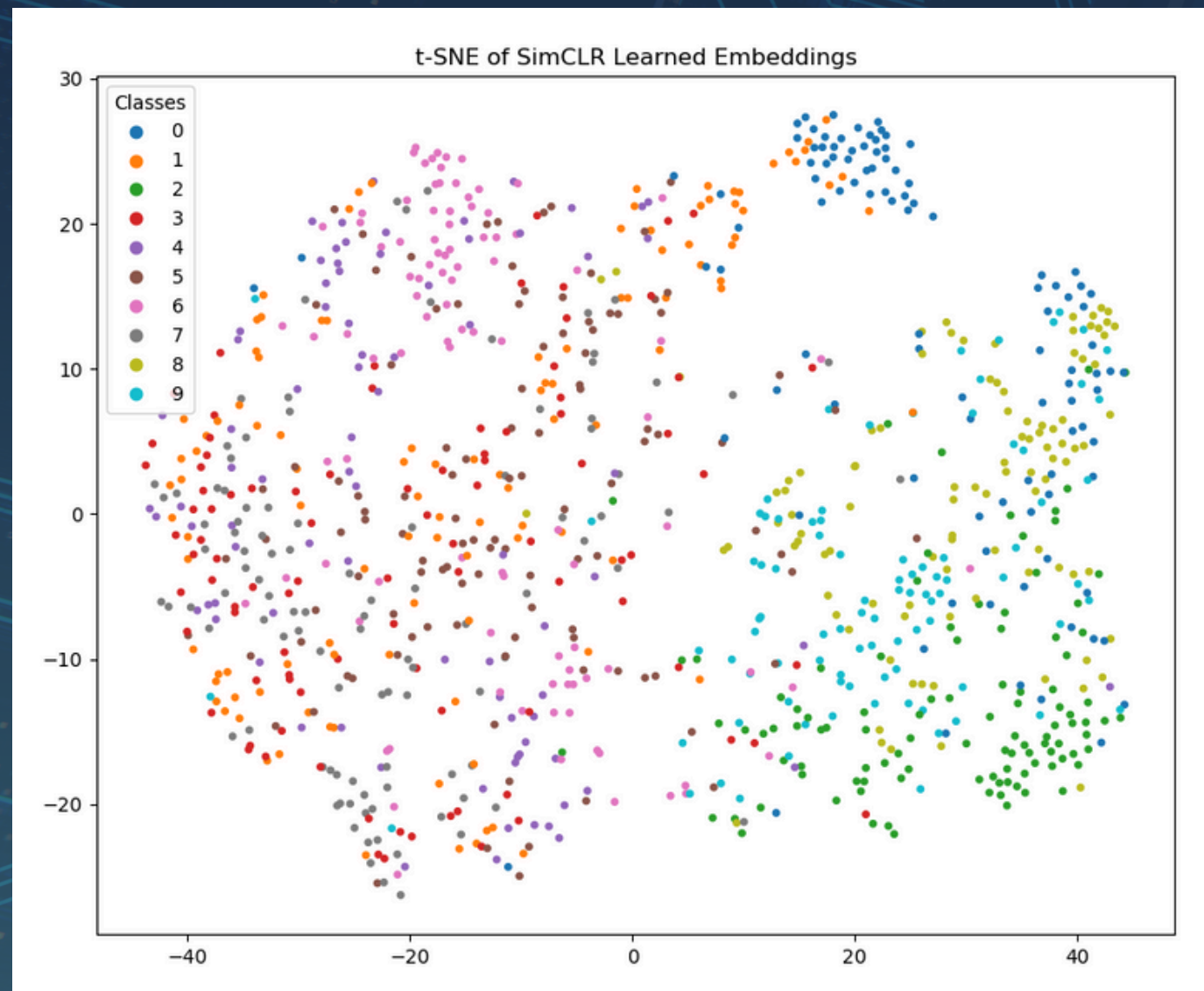
ENVIRONMENT SETUP

- Chose STL-10 dataset (100K unlabeled + 13K labeled)
- Built training pipeline using PyTorch + torchvision
- Customized SimCLR-style augmentations for contrastive pairs





PROGRESS CONTD...



3

SIMCLR IMPLEMENTATION

- ResNet-18 encoder + MLP projection head
- Trained on 100K unlabelled images of STL-10 for 10 epochs on the MacBook Pro M2 silicon chip.
- Used NT-Xent loss for contrastive objective

4

INITIAL RESULTS

- t-SNE shows clustering of semantically similar images
- Linear evaluation (classifier on frozen embeddings):
- 55.24% accuracy on labeled validation set after 10 epochs of training



WORK LEFT...

1.

EXTEND MODELS

- Implement MoCo and BYOL
- Modularize codebase for clean switching between models

2.

EVALUATION & COMPARISON

- Perform linear probing and t-SNE/UMAP visualization
- Benchmark all models vs. supervised baseline

3.

OPTIMIZATION & REPORTING

- Tune augmentations, batch size, temperature
- Finalize report with results, analysis, and GitHub code



**THANK
YOU!**